3.30 Report

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March 30, 2026

1 Last Week's Experiments

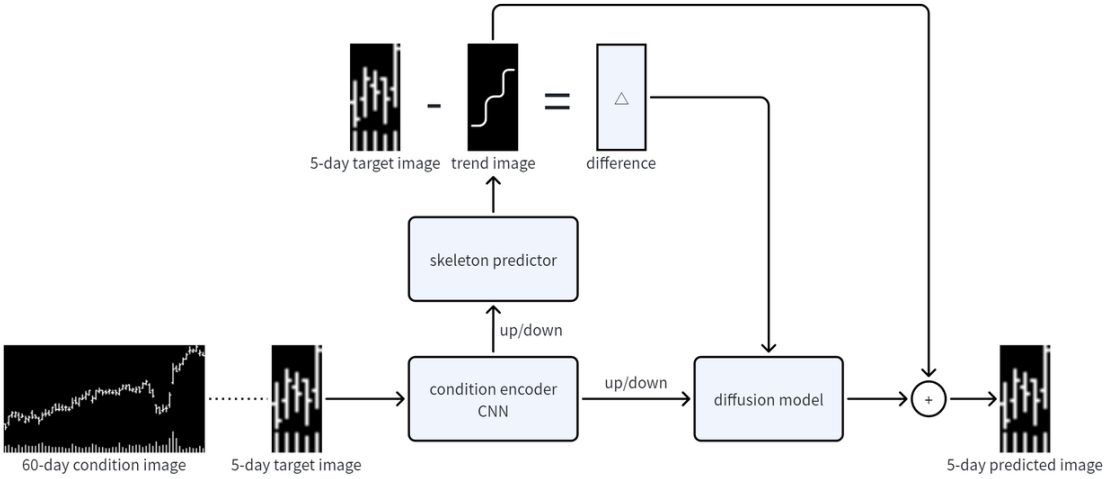

Figure 1: Updated pipeline overview with all modifications applied

1.1 Modifications Implemented

The following six changes were applied to the pipeline based on last week's plan:

1. **Price / Volume separation**: The target is simplified to price-only prediction, excluding volume generation. This reduces the output dimensionality and focuses the model on the most critical signal.
2. **CNN generates data instead of images**: The CNN now outputs raw numerical data rather than rendered chart images. Chart rendering is deferred to a later stage inside the diffusion model, eliminating pixel-level artifacts from direct image regression.
3. **Shared CNN Condition Encoder for Skeleton Predictor**: The CNN-based condition encoder now also feeds into the Skeleton Predictor, providing a richer conditioning signal for coarse-level prediction.
4. **Cross-attention in Skeleton Predictor**: A cross-attention mechanism is introduced where
   - **K, V** = historical 60-day token embeddings
   - **Q** = 5 future-day query tokens

   Each future-day token independently attends to the full 60-day history, extracting the most relevant information for its own prediction.

5. **Fused Condition Encoder and Skeleton Predictor**: Both modules are CNN-based. The condition encoder is pre-trained and then frozen; one copy of its output serves as the diffusion condition guide, while another copy is jointly fine-tuned with the Skeleton Predictor to adapt the shared features for coarse prediction.

6. **Augmented loss with pattern labels**: Inspired by **YOLO**-style detection, explicit pattern recognition objectives are added to the loss. The pattern labels are extracted from the historical 60-day chart image:

   - **Slope**: regression target for local price gradient
   - **Direction**: binary up/down classification
   - **Shape**: categorical label for candlestick pattern type (e.g., hammer, engulfing)

1.2 Results

1.2.1 Sample Output

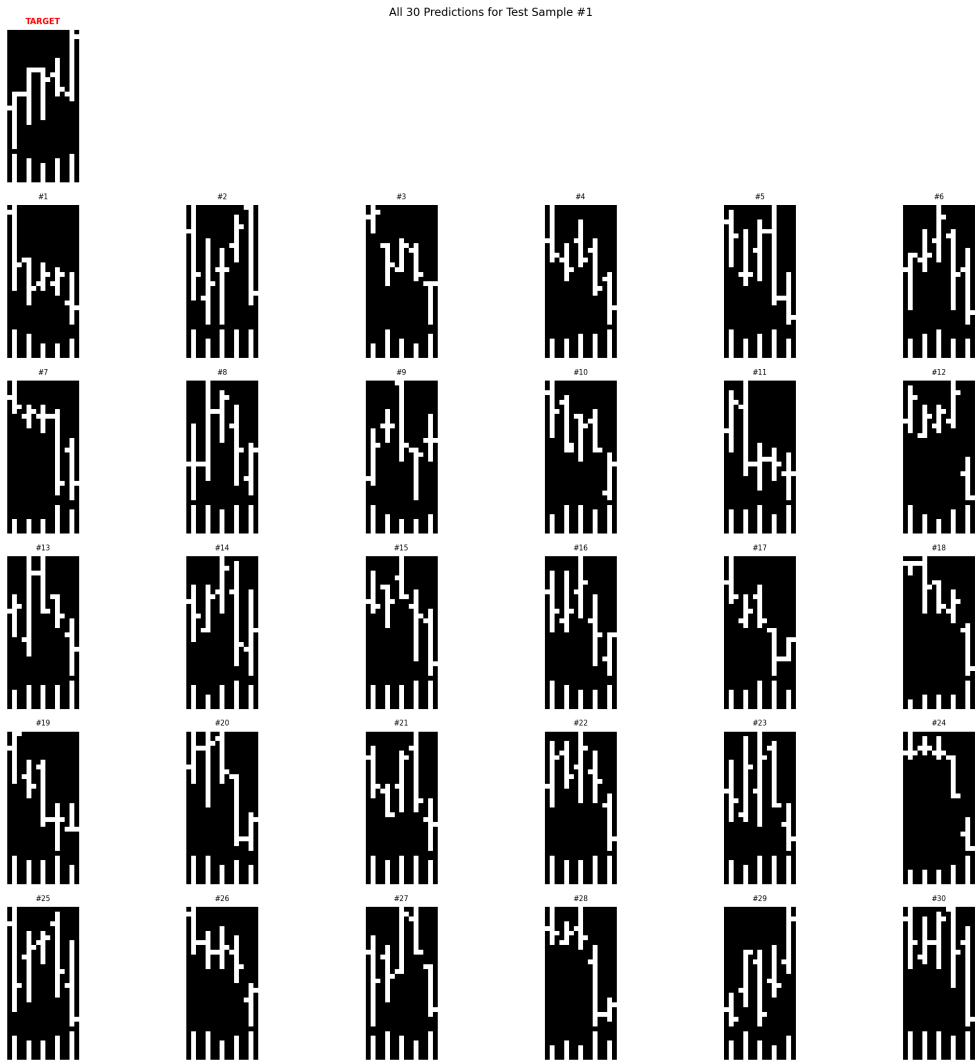

Figure 2: Generated sample: skeleton prediction and diffusion refinement with the updated pipeline

1.2.2 Portfolio Performance

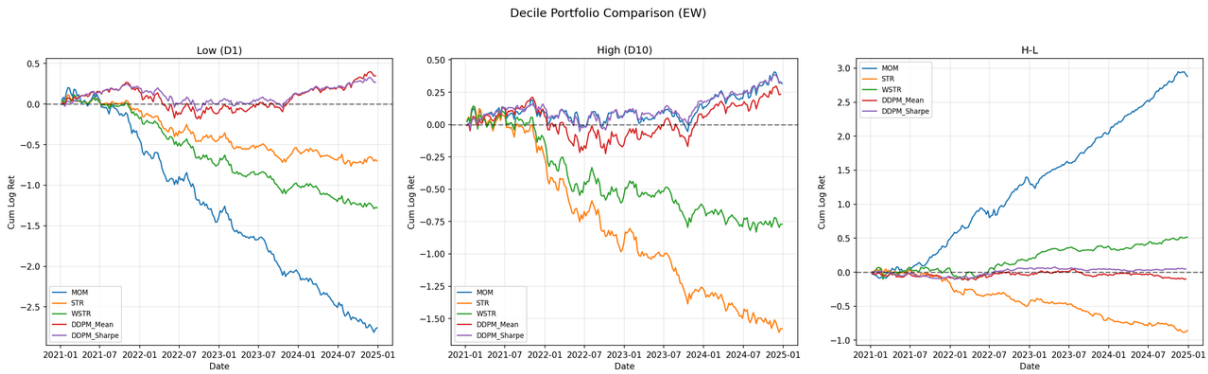

Figure 3: Portfolio result (1)

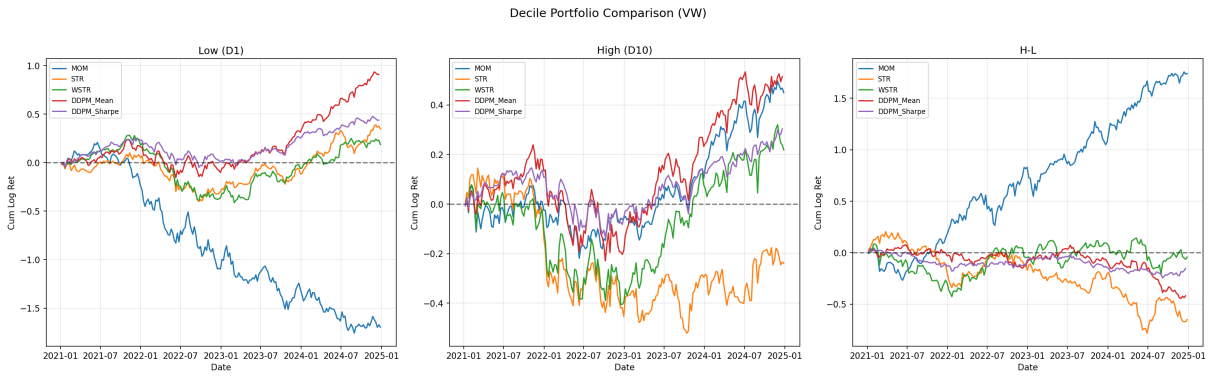

Figure 4: Portfolio result (2)

Table 4: Portfolio Performance Restricted to 500 Largest Stocks

	Equal Weight													
	I5/R5		I20/R5		I60/R5		MOM/R5		STR/R5		WSTR/R5		TREND/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	0.05	0.28	0.06	0.35	0.03	0.16	0.07	0.23	0.09	0.37	0.10	0.42	0.07	0.26
2	0.06	0.34	0.06	0.36	0.04	0.24	0.09	0.39	0.12	0.60	0.11	0.61	0.07	0.35
3	0.07	0.39	0.07	0.41	0.06	0.33	0.10	0.54	0.11	0.64	0.12	0.65	0.09	0.47
4	0.09	0.50	0.10	0.52	0.10	0.51	0.10	0.63	0.11	0.68	0.12	0.78	0.11	0.66
5	0.07	0.41	0.08	0.45	0.08	0.45	0.11	0.72	0.10	0.66	0.09	0.57	0.11	0.67
6	0.10	0.54	0.09	0.47	0.09	0.51	0.10	0.72	0.10	0.65	0.11	0.68	0.11	0.74
7	0.08	0.42	0.08	0.45	0.11	0.57	0.11	0.72	0.10	0.60	0.10	0.65	0.10	0.67
8	0.10	0.53	0.11	0.56	0.11	0.60	0.09	0.60	0.09	0.52	0.09	0.53	0.10	0.68
9	0.13	0.68	0.12	0.61	0.13	0.69	0.09	0.50	0.08	0.47	0.08	0.46	0.10	0.63
High	0.17	0.82	0.15	0.75	0.14	0.79	0.11	0.53	0.07	0.32	0.05	0.21	0.11	0.51
H-L	0.12***	1.02	0.09***	0.78	0.11***	1.08	0.05	0.19	-0.02	-0.11	-0.05	-0.30	0.04	0.22
Turnover	712%		676%		638%		30%		47%		47%		42%	
	Value Weight													
	I5/R5		I20/R5		I60/R5		MOM/R5		STR/R5		WSTR/R5		TREND/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR		
Low	-0.01	-0.05	0.03	0.18	0.02	0.13	0.06	0.20	0.08	0.33	0.09	0.39	0.08	0.34
2	0.04	0.21	0.04	0.23	0.03	0.18	0.08	0.38	0.10	0.53	0.08	0.46	0.07	0.32
3	0.06	0.33	0.06	0.31	0.05	0.26	0.10	0.57	0.10	0.58	0.10	0.64	0.09	0.52
4	0.09	0.48	0.06	0.33	0.08	0.43	0.12	0.72	0.11	0.70	0.11	0.73	0.09	0.57
5	0.06	0.32	0.07	0.40	0.05	0.26	0.08	0.52	0.11	0.67	0.07	0.47	0.11	0.70
6	0.09	0.52	0.07	0.41	0.08	0.45	0.09	0.67	0.08	0.59	0.10	0.66	0.10	0.69
7	0.07	0.37	0.09	0.53	0.09	0.49	0.08	0.58	0.09	0.55	0.10	0.69	0.08	0.49
8	0.11	0.61	0.11	0.56	0.08	0.47	0.08	0.56	0.08	0.50	0.08	0.49	0.09	0.59
9	0.13	0.69	0.12	0.63	0.10	0.57	0.08	0.46	0.08	0.42	0.07	0.38	0.08	0.51
High	0.16	0.80	0.15	0.77	0.15	0.83	0.11	0.53	0.06	0.25	0.05	0.23	0.08	0.40
H-L	0.17***	1.29	0.12***	0.96	0.13***	1.03	0.06	0.23	-0.02	-0.10	-0.04	-0.19	-0.00	-0.02
Turnover	722%		693%		657%		29%		48%		734%		575%	

Figure 5: Portfolio result (3)

1.2.3 Value at Risk (VaR)

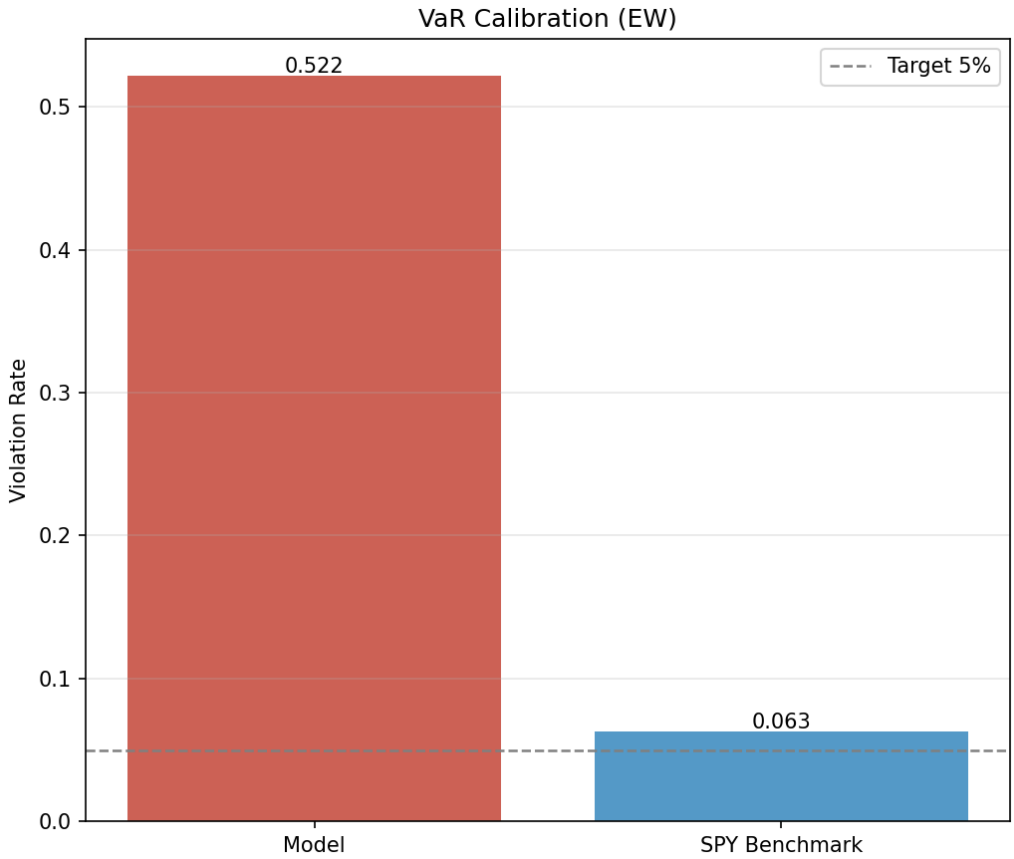

Figure 6: Value at Risk evaluation (1)

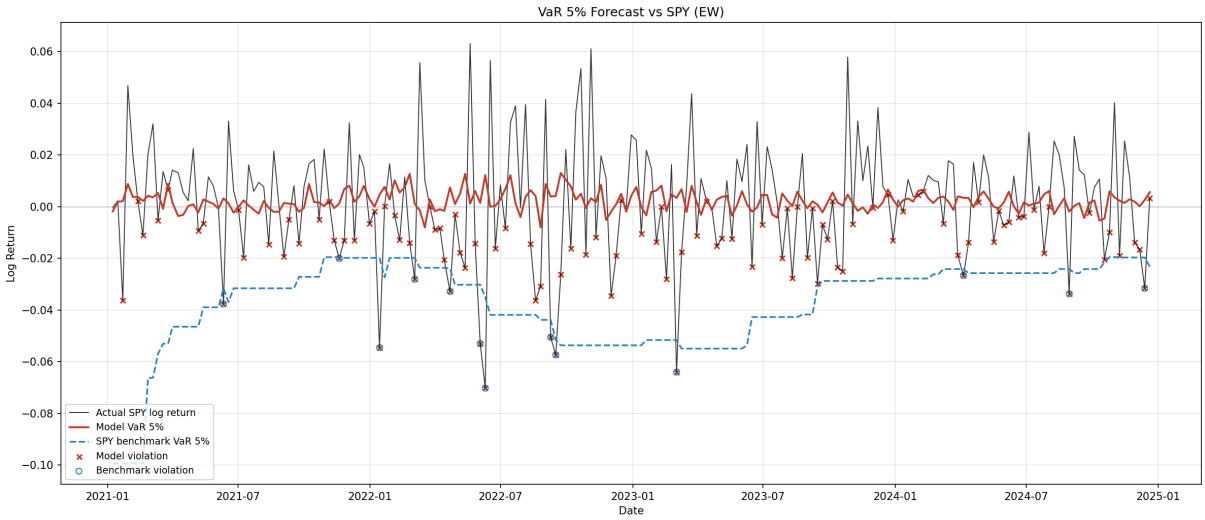

Figure 7: Value at Risk evaluation (2)

2 Model Modification

2.1 Architecture

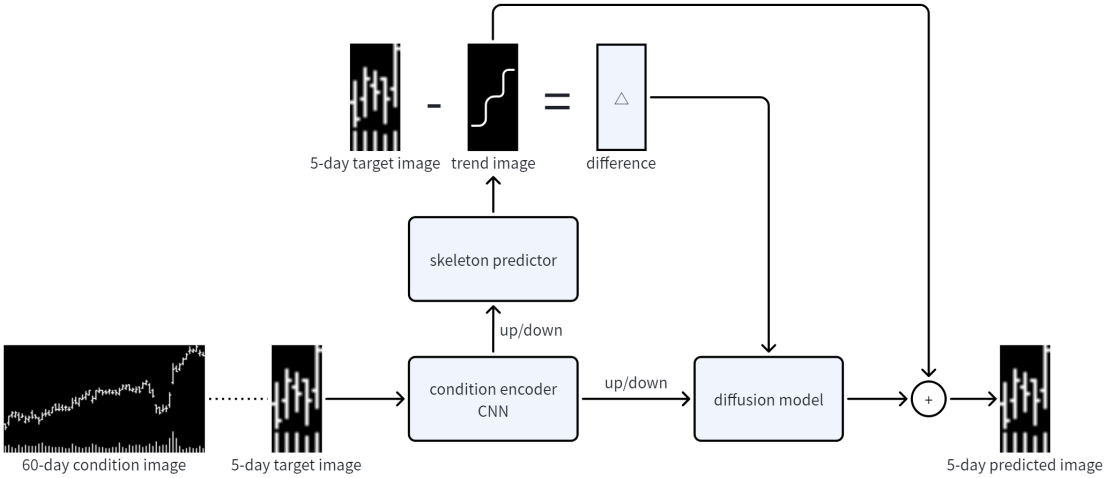

Figure 8: Previous model architecture

2.2 Sample Output

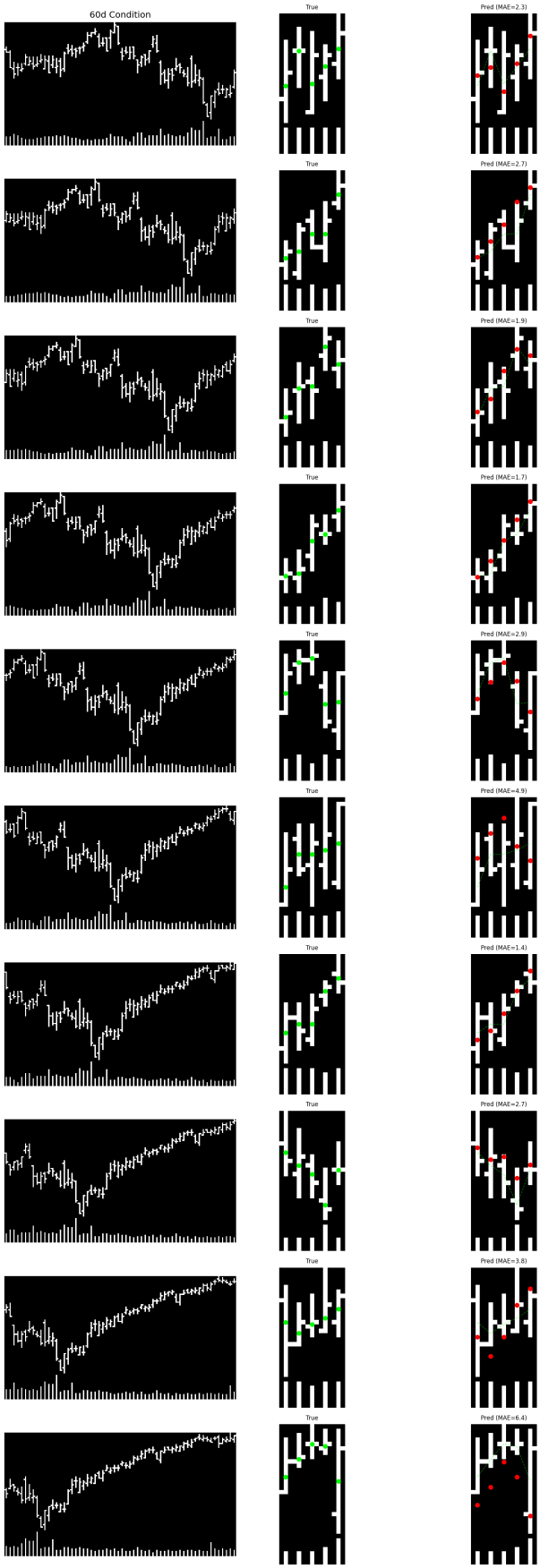

Figure 9: Sample output from the revised model (1)

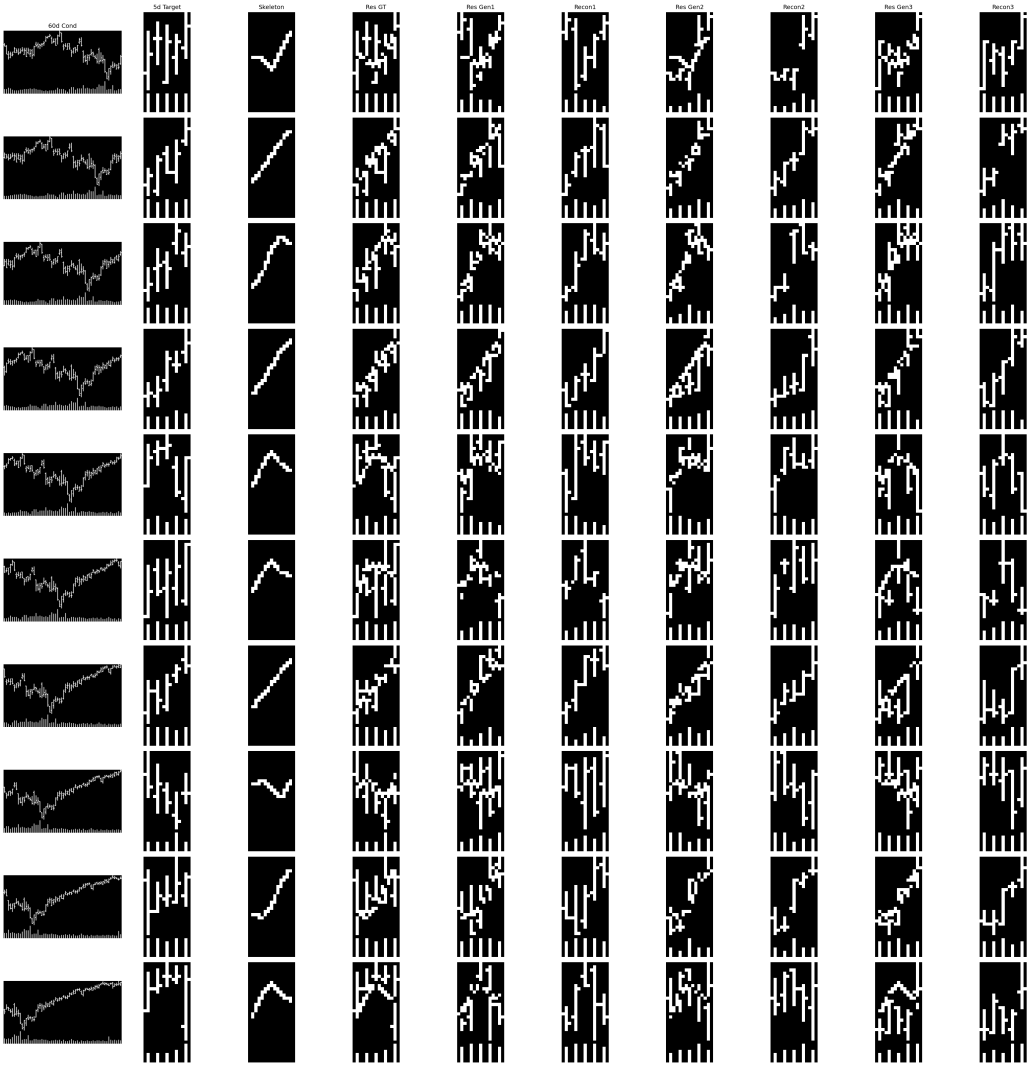

Figure 10: Sample output from the revised model (2)

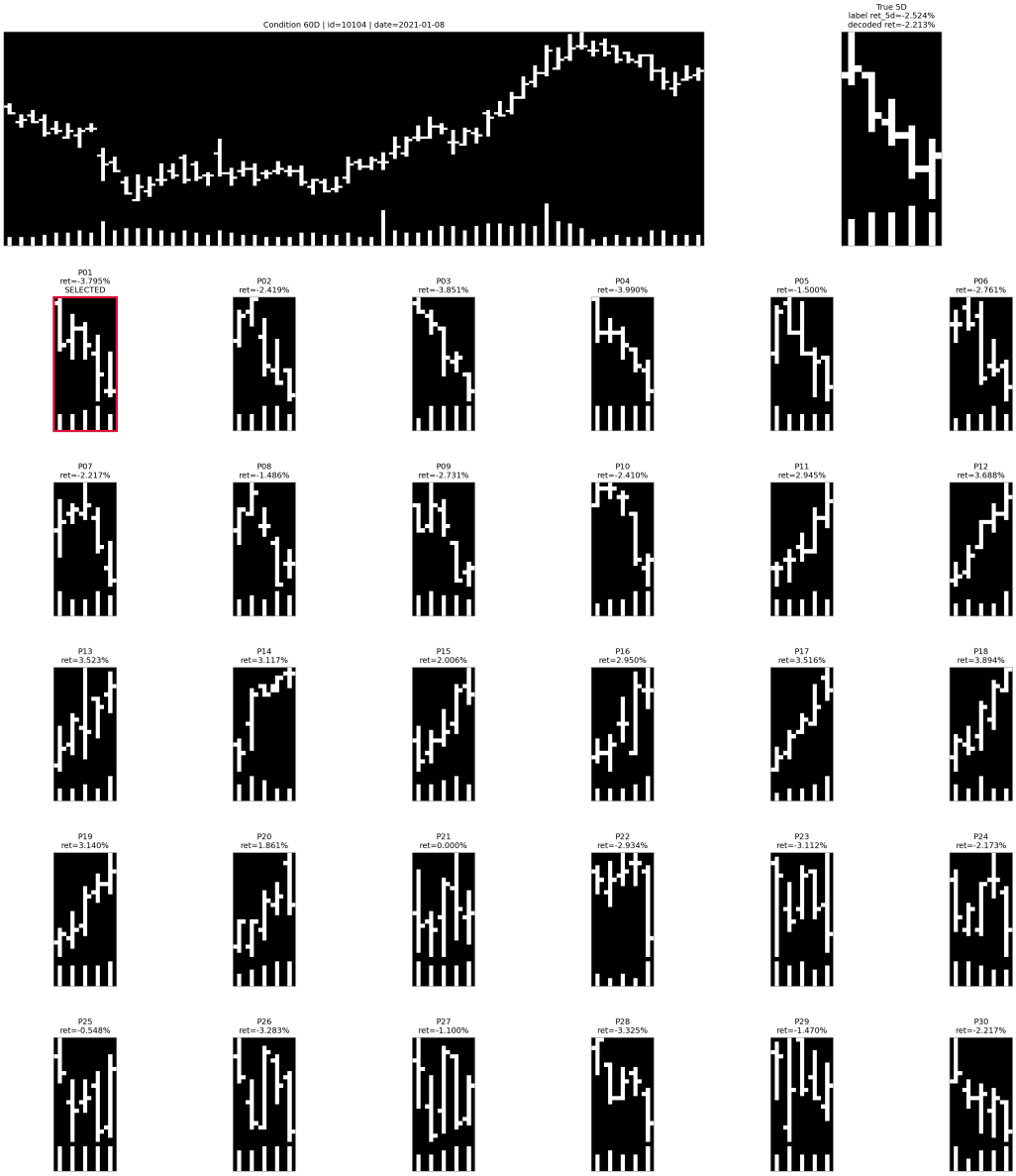

Figure 11: Sample output from the revised model (3)

2.3 Results

2.3.1 Portfolio Performance

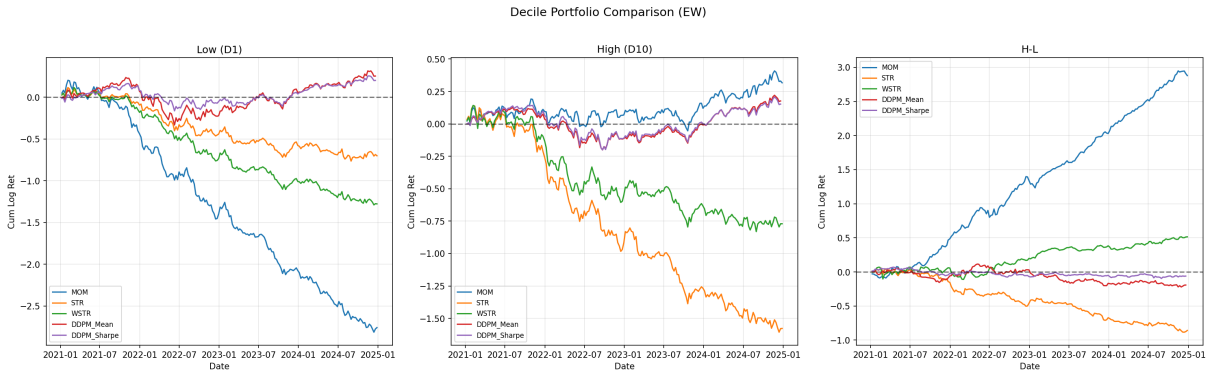

Figure 12: Portfolio backtest — revised model (1)

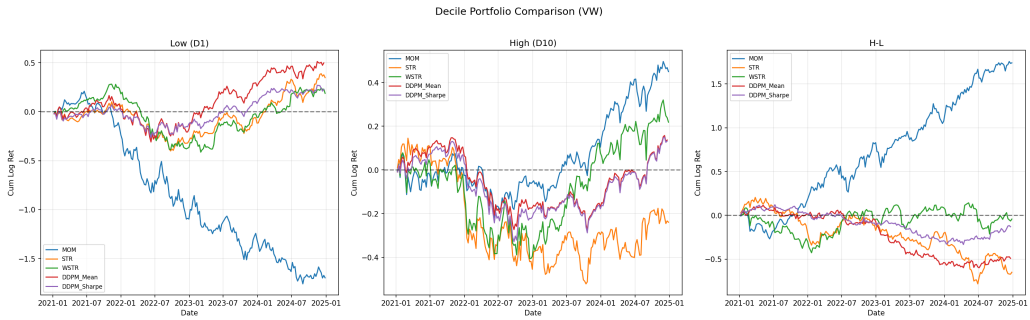

Figure 13: Portfolio backtest — revised model (2)

2.3.2 Value at Risk (VaR)

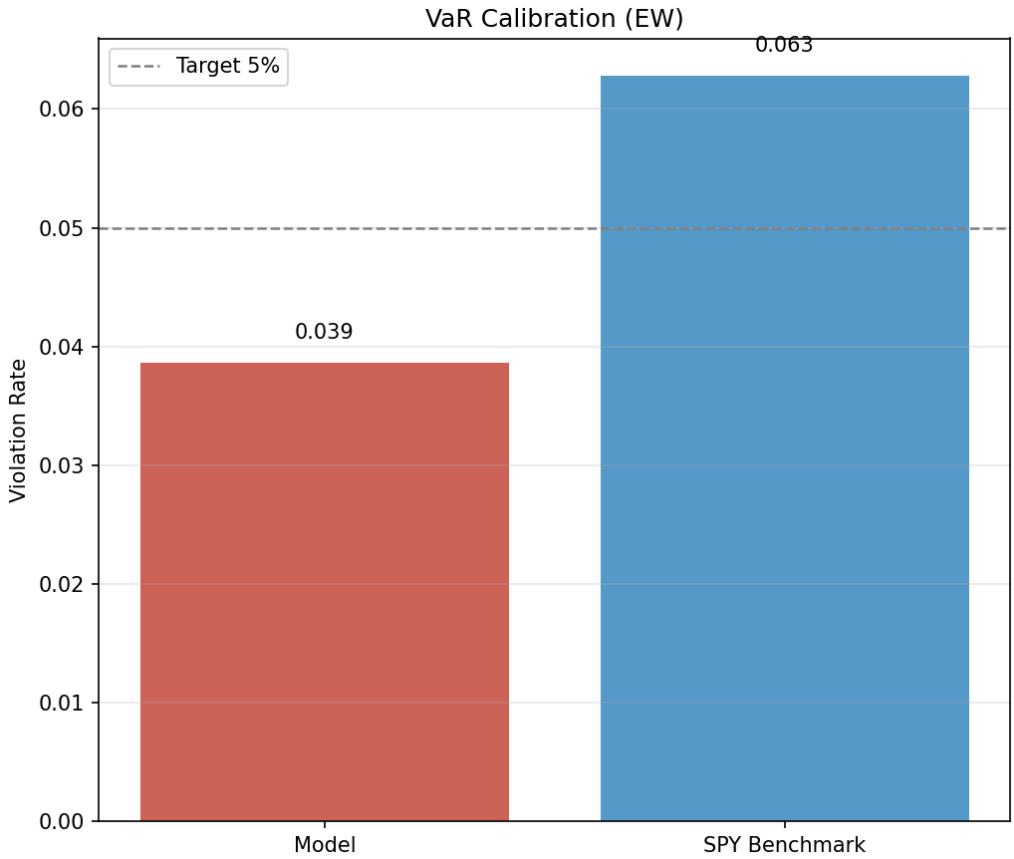

Figure 14: Value at Risk — revised model (1)

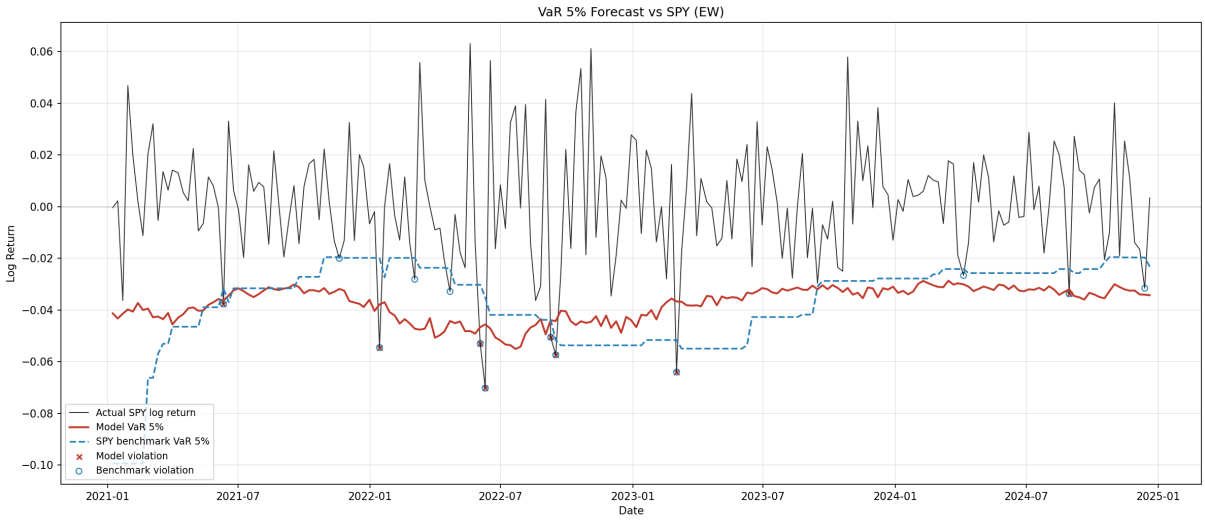

Figure 15: Value at Risk — revised model (2)

2.4 Analysis: Why Do Samples Look Good but Accuracy and Portfolio Remain Poor?

The root cause lies in the **top-k pattern aggregation** strategy used during inference:

1. The pipeline first retrieves the **top-k most similar historical patterns** from the pattern library.
2. For each retrieved pattern, the model generates a skeleton prediction and a diffusion sample independently.
3. All k candidate outputs are then merged and averaged to produce the final prediction.

Problem: When the top-k set contains patterns with conflicting directional signals (e.g., one bullish pattern and one bearish pattern), their averaged output cancels out the directional information. This leads to a "blurred" mean chart where the predicted direction can even *oppose* the dominant signal — explaining why individual samples appear visually plausible, yet the aggregate prediction yields poor directional accuracy and weak portfolio performance.

Solution: Switching from **top-k** to **top-1** eliminates the cross-pattern mixing entirely, ensuring the final output preserves the directional commitment of the single best-matching pattern.

3 Next Steps

1. **Switch from top-k to top-1**: Remove the cross-pattern averaging to preserve directional signal integrity, as identified in the analysis above.
2. **Increase pattern diversity**: Expand the pattern library with more pattern categories to improve coverage of market conditions and enhance the skeleton predictor's ability to discriminate between distinct future trajectories.
3. **Add VaR baseline models**: Introduce standard risk-modeling baselines for Value at Risk comparison:
   - **GARCH**: parametric volatility model as the classical benchmark
   - **Neural Network (NN)**: feed-forward network baseline
   - **LSTM**: recurrent network baseline capturing sequential dependencies